

Comparison of the Effects of Walking and Weight Training Programs on Body Image in Middle-Aged Women: An Experimental Study

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Abstract

Purpose. This study was conducted to determine the effect of a resistive training intervention on body image in middle-aged women compared to an exercise walking program. Another purpose was to develop two multivariate models to explain improvement in body image among the lifters and walkers.

Design. A pretest-posttest experimental design with random assignment of subjects to two exercise groups was employed.

Setting. Subjects trained in their own homes and were tested at the university.

Subjects. Participants were 60 women recruited from the local community with an average age of 42.5 ± 4.2 years.

Intervention. Subjects were randomly assigned to a resistive training or exercise walking program, both of which were three days per week and 12 weeks in duration.

Measures. Body cathexis was assessed using the Body Cathexis Scale; cardiovascular endurance was measured using the one-mile walk test, and muscular strength was assessed using standard weight training procedures.

Results. After completion of the exercise interventions, lifters showed greater muscular strength than walkers, and walkers displayed greater cardiorespiratory endurance than lifters, as expected. Lifters also improved significantly more in body image than the walkers.

Conclusions. Participation in a three-day-per-week resistive training program seems to improve body image in middle-aged women more than participation in a three-day per week walking program. Further, it appears that hard work and fitness improvements contribute significantly toward positive changes in body image. (*Am J Health Promot*, 1993; 8(1):34-42)

Key Words: Body Cathexis, Fitness, Exercise, Resistive Training, Psychological Health

INTRODUCTION

Health professionals frequently encourage exercise to enhance well-being. The American College of Sports Medicine (ACSM) recommends that healthy adults engage regularly in aerobic and resistive training— aerobic activity to enhance cardiorespiratory fitness and assist in weight control, and resistive work to build muscular strength and endurance.¹ However, the benefits of exercise are not limited to physical health. Regular exercise may be helpful in buffering the effects of stress, anxiety, and depression.^{2,5} Furthermore, several literature reviews suggest that psychological well-being can be improved as a consequence of exercise.⁶⁻¹¹

Most studies that have examined the effect of exercise on psychological well-being have employed aerobic activities to engender change.⁶⁻¹¹ Relatively few have focused on resistive exercise or weight training. Yet, according to Tucker,¹² "weight training may be superior to aerobic activity for improving psychological health because of the noticeable physical changes which tend to result from regular training," and because "the training environment contains an abundance of positive feedback as the need to increase progressively the training load to match the body's increased strength can be seen and is often recorded by the lifter."¹³

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Presently, very little research is available showing whether or not weight training actually is superior to aerobic exercise for improving emotional health. Several years ago, Trujillo¹⁴ compared the effects of a jogging program and a weight training intervention on college women and found that both programs resulted in significant increases in self-esteem. However, the weight training program had a significantly greater effect on the women's self-esteem than the jogging program. Unfortunately, this study employed only female college students as subjects greatly reducing the generalizability of the findings, and an experimental design was not used; hence, cause-and-effect conclusion are not warranted.

Most research conducted to examine the effect of weight training on psychological health has used passive control groups, and the majority of studies have focused on male subjects.^{13,15-19} Relatively few have studied the resistive training/psychological health association using female participants,^{12,14,20,21} even though weight training among females is rapidly gaining popularity.²²

To date, only one investigation by Brown and Harrison²⁰ has studied the psychological effect of weight training on females using an experimental design. Findings showed that muscular strength and self-esteem were enhanced as a result of the 12-week weight training program.

In 1992, Tucker and Maxwell¹² studied the effects of weight training on the emotional well-being of 152 females. A multivariate model helped to explore the factors which contributed to the emotional health changes of the subjects in the weight training program, such as improvements in muscular strength, reduced skinfold measures, and decreased body weight. Although significant improvements in emotional well-being associated with the weight training intervention were revealed, deficiencies of this study were failure to randomly assign subjects to groups and use of only college females as subjects.

Collectively, studies have exam-

ined many facets of the exercise and emotional health relation, but very few investigations have used experimental designs, still fewer have included multivariate models, and most have employed men as subjects, frequently college students. Hence, the present study was designed to improve on these shortcomings. Its purpose was to determine the effect of a resistive training intervention on emotional well-being, particularly body cathexis, compared to an exercise walking program using an experimental design and middle-aged women as subjects. A secondary purpose was to develop two multivariate models to explain the variation in improvement in body cathexis among participants in the lifting and walking exercise programs. Another ancillary purpose was to determine the effects of regular resistive training and walking on dynamic and absolute strength and cardiorespiratory endurance in middle-aged women.

METHODS

Design

A pretest-posttest experimental design was employed in the present study with subjects randomly assigned to participate in either a resistive training program or a walking program three days per week. Demographic, physical, psychological, and performance variables were assessed that might be predictive of changes in body cathexis in the lifters and the walkers. A passive control group was not used in the study for two primary reasons. First, research has already shown multiple times that resistive training improves emotional well-being in women, particularly body image, compared to passive controls.^{12,20,21} Second, sedentary women in the community who volunteer to participate in fitness-related research projects usually do so because they desire to begin training and improve their fitness levels. When volunteers are asked to remain sedentary, yet perform all of the other duties and work required of subjects for many weeks, they tend to become disappointed and attrition frequently runs high.

Lack of a passive control group does not affect conclusions that one type of exercise, such as resistive training, is superior to another, such as walking, for enhancing body image. However, without passive controls it is not valid to conclude that all of the psychological change that accompanies an exercise intervention is a function of the intervention.

Sample

A total of 65 sedentary, nonobese women between 35 and 49 years of age were selected from individuals who responded to an advertisement in a newspaper with subscribers in cities throughout Utah County, Utah. Each volunteer completed a medical screening questionnaire and only those who reported freedom from smoking, coronary heart disease, diabetes mellitus, hypertension, and recent surgery were allowed to participate. Only women who were less than 120% of ideal weight according to the Metropolitan Life height-weight table were used as subjects to control for the confounding effects of obesity. All subjects signed an informed consent form approved by the University Internal Review Board. During the course of the 12-week project, five of the 65 women dropped out of the study for various reasons, including divorce, transportation problems, illness, employment change, and family move, leaving 30 women in each of the two exercise groups. Those who left the study did not differ from the other women on any of the pretest variables.

At the onset of the study, the 60 subjects averaged 42.5 ± 4.2 years of age (lifters: 41.6 ± 3.9 , walkers: 43.4 ± 4.3), weighed 140.0 ± 21.6 pounds (lifters: 138.0 ± 19.7 , walkers: 142.0 ± 23.3) and were 64.8 ± 2.2 inches tall (lifters: 65.2 ± 2.2 , walkers: 64.5 ± 2.3). Approximately 97% of the women were white, 88% were married, and 45% reported having had some college education. None of the subjects were regular exercisers at the onset.

Measures

Body Cathexis. The principal

dependent variable of this study was body cathexis or body image, which reflects the feelings individuals have toward their bodies, positive or negative. Body cathexis was employed in the present investigation because of its well established validity and empirical links with exercise,²³ somatotype and body build,^{18,24-26} and resistive training in men¹⁶⁻¹⁹ and women.¹²

The Body Cathexis Scale is viewed as one of the best inventories for measuring body image.^{27,28} Participants evaluate the appearance of selected body parts, such as arms, legs, shoulders, waist, and hips, and various functional and/or physical constructs, such as stamina, muscular strength, flexibility, tolerance for pain, and body fatness, on a five-point Likert scale ranging from "feel very positive about" to "feel very negative about." The scale contains a total of 40 items with possible scores ranging from a low of 40 to a maximum of 200. Psychometric properties and principal components of the Body Cathexis Scale have been thoroughly researched.²⁸⁻³⁰

The same version of the Body Cathexis Scale used by Tucker and Maxwell¹² was employed in the present investigation. This version of the instrument was very similar to those used by Tucker in other exercise and body image studies,^{13,17-19,24-26,28-30} only more information was added to a few items. For example, originally one item for subjects to rate was "arms." Pilot research showed that the single word "arms," was a bit vague and confusing. Should subjects interpret "arms" as appearance of arms or strength of arms? Hence, this and similar items were clarified to help subjects know better what they were evaluating. "Arms" was changed to two items, "appearance of arms," and "strength of arms."

The version of the Body Cathexis Scale used in this study was administered to a group of 30 women in a pilot test to ascertain test-retest reliability with one week between the tests. Results showed no significant difference between the test and retest means, and the Pearson

correlation was .90.

Muscular Strength. Dynamic and absolute (1 RM) strength were assessed by the researchers with help from two trained research assistants. Dynamic strength scores were generated for all subjects using the Universal Gladiator Gym by recording the maximum number of repetitions completed using 45% of body weight in the bench press and 140% of body weight in the leg press,^{31,32} with knees flexed at 55 degrees. "Total dynamic strength" was determined by adding the scores of the two exercises.

Absolute strength scores (1RM) were determined by recording the maximum amount of weight each woman could lift for one repetition in the bench press, leg press, seated behind-neck press, and arm curl.³¹ The Universal Gladiator Gym was used to assess absolute strength in the bench press and the leg press, whereas free-weight barbells were employed to measure absolute strength in the seated behind-neck press and the arm curl. One repetition maximums (1RM) were determined using the "trial-and-error" method.³³ "Total absolute strength" was determined by summing the scores of the four lifts.

Height and weight were measured with subjects wearing gym shorts, t-shirts, and no shoes, using a standard balance scale that included a sliding gauge for height.

Cardiovascular Endurance. The One-Mile Walk Test was used to measure cardiorespiratory endurance as recommended by the American College of Sports Medicine.³¹ The test protocol required each subject to warm up and then walk one mile as fast as possible. Only walking was allowed. Performance time was used as an aerobic fitness score for each subject.

Subjective Ratings. At the conclusion of the study, subjects were asked to rate their feelings on four items arranged on seven-point Likert scales asking: 1) how hard

they feel they had trained during the 12-week intervention ranging from mildly to extremely hard (perceived intensity); 2) their current physical fitness level ranging from very poor to excellent); 3) the extent to which they feel their fitness improved during the study, ranging from minimally improved to extremely improved; 4) the extent they enjoyed their workouts, ranging from greatly disliked to greatly enjoyed.

At the conclusion of the study, subjects also reported: 1) whether or not they planned to continue exercising, and if so, 2) whether they planned to weight train, 3) walk, and 4) whether or not they planned to participate in some form of exercise other than resistive training or walking.

Intervention

In an introductory meeting, the women volunteers were briefly oriented regarding the study. All of the subjects were given a medical screening questionnaire during the first meeting to assess risks that might prevent their involvement in the study. Those who possessed unacceptable risk factors were dismissed from the study. Subjects also completed a written questionnaire that included general demographic items, lifestyle questions, and the Body Cathexis Scale.

All subjects met a second time during the next week at their convenience and were pretested on dynamic and absolute strength, and on a separate day, cardiovascular endurance. After the pretests, subjects were randomly assigned to the resistive training or walking group.

At the conclusion of the 12-week exercise intervention, subjects again completed the strength and cardiorespiratory fitness tests, the Body Cathexis Scale, and other questions regarding their recent exercise experience and future exercise plans.

Resistive Training Program. A meeting was held a few days after the pretests were completed for the resistive training group only. These women were given a demonstration

and trained regarding the exercises they would be required to execute in their regular workouts. All of the subjects in the resistive training group were given a Body Bar to use during the study. The Body Bar is a resistive training device that consists of a padded bench and a bar. The bar is secured at the head of the bench near the floor with two elastic cords, one attached to each end of the bar. Resistance can be altered by twisting the bar thus having the two elastic cords wrap around the ends of the bar. As the cords become shorter, resistance is increased and vice versa. A number of resistive training exercises can be performed using the Body Bar.

Subjects in the resistive training group were informed that they should train three times per week in their homes using the Body Bar, preferably on Mondays, Wednesdays, and Fridays. They were given explicit instructions regarding how to record their results, and they were also asked to report in either by telephone or in person after each workout session. Contact after each workout was required to keep compliance as high as possible.

During the resistive training group meeting, the women were given a specific written resistive training program to follow. Each woman was to execute the same program. The complete program consisted of warm-up exercises, the number of sets and repetitions that should be performed, the order of the exercises, the recovery time between sets, and the intensity of the workout.

After subjects were warmed-up, they were to proceed to the resistive training activities which included six exercises; bench press, seated behind-neck press, arm curl, seated rowing, squat, and abdominal curl ups. Three sets of 10 repetition maximums were to be performed for each exercise, except during the first two weeks of training (six workouts), when 10 mild-moderate repetitions were to be performed for each resistive exercise. The women were instructed to perform five to 20 abdominal curl-ups per set, depend-

ing on their fitness level.

Intensity of each set was determined by the number of repetitions subjects felt they could complete beyond the assigned 10 repetitions. If subjects felt that no repetitions could be executed immediately following the assigned 10 repetitions, then the set was considered high intensity. If they felt one or two repetitions could be completed following the assigned 10 repetitions, then the set was considered medium intensity, and if subjects felt three or more repetitions could be completed beyond the assigned 10 repetitions, then the set was considered low intensity. The intensity of each set of each lift was recorded by those in the resistive group.

The six exercises were paired into three groups. Group one, squat and seated behind-neck press; group two, seated rowing and curl-ups; and group three, bench press and arm curl. The women were to perform the two exercises in group one first (squat and seated behind-neck press). Each exercise was to be alternately performed with the exercise in which it was paired to reduce the amount of time for the workout while allowing time for the just-exercised muscle group to recover. Subjects then completed the same process for the exercises in group two (seated rowing and curl up) and then the group three exercises (bench press and arm curl). The guidelines indicated that there should be 30 to 90 seconds of recovery between each set. If at any time during an exercise on the Body Bar intensity or resistance was too low or too high, the subject was to quickly change the resistance by twisting the bar to shorten or lengthen the elastic cords and increase or decrease resistance.

During each workout, subjects recorded on special forms details regarding each activity, particularly the number of repetitions and intensity of each set. At the conclusion of each workout, subjects recorded how they felt overall on a Likert scale ranging from excellent to very poor.

Walking Program. Another meeting was held for only subjects in the

walking group during the time of the resistive training meeting. At this meeting, women in the walking group were given an explicit walking program. The subjects were instructed to walk three times per week, preferably on Mondays, Wednesdays, and Fridays, covering a distance of one mile per session during the first three weeks, two miles per workout during weeks four through eight, and two and a half miles per session during weeks nine through 12.

The walkers were asked to work at an intensity of 65% of their estimated maximum heart rates for the first three weeks of the program, increase to 70% for the next five weeks, and then increase to 80% for the remaining four weeks of the 12-week training period. During the meeting, walking subjects were trained regarding the proper technique for assessing exercise heart rate.

Walking group subjects were instructed to record their walking activities after each session on a special log, including their walking times and distances, exercise heart rates, and how they felt overall on a six-item scale ranging from excellent to very poor.

Weekly Seminars. During each of the 12 weeks of training, an hour-long evening meeting was held for all subjects, both from the resistive training and walking programs. At this time, the women turned in their training logs and asked questions. During each meeting there was a speaker who addressed a health or fitness topic. The purpose of the weekly meetings was to keep subjects participating and informed and to receive feedback to minimize workout problems.

Data Analysis

Multiple regression using the General Linear Model (GLM) strategy in the SAS/STAT software program (Cary, North Carolina) was used to test between-group differences in muscular strength, cardiovascular endurance, and body cathexis. Partial correlation was employed to determine the extent to

which the lifters and walkers differed after controlling for potential confounding factors, particularly pretest differences. To develop a profile of the women who tend to benefit most psychologically as a result of resistive training or walking, multiple regression was used to determine the degree to which selected potential mediating variables accounted for changes observed in body cathexis across the 12-week study. The sequence in which the predictor variables were entered in the regression models was first objective performance scores, followed by objective physical variables, and lastly subjective psychological measures.

RESULTS

There were no significant differences on the pretest between the randomly assigned women who trained resistively and those who walked regarding age, height, weight, marital status, or education, nor any of the baseline performance scores, except the walking test, in which lifters scored better than the walkers. After 12 weeks of training, lifters showed significantly greater improve-

ments in muscular strength, dynamic and absolute, compared to the walkers, as depicted in Table 1. Among the 54 subjects who completed at least 80% of the assigned workouts across the 12-week study, women in the resistive training group increased 145.8% in total dynamic strength and 11.5% in total absolute strength, compared to increases of 61.2% in total dynamic strength and 3.6% in absolute strength for the walkers across the three-month intervention. Women in the resistive group improved 2.4 times more in dynamic strength and 3.2 times more in absolute strength compared to women in the walking group. Walkers improved an average of 83.5 ± 64.5 seconds in the one-mile walk (unadjusted), whereas the lifters improved 44.5 ± 53.5 seconds. Women in the walking group improved their one-mile walk times 87.6% more than the lifters during the 12-week intervention. Additionally, walkers reported an average exercise heart rate of 125.6 ± 8.4 beats per minute during their walking sessions, and their mean walking time or training duration was 29.2 ±

2.1 minutes over the 12 weeks. There were no significant changes in body weight within or between the two training groups across the exercise intervention. The resistive training group began the study with an average body weight of 138.0 ± 19.7 pounds and the walkers commenced with a mean of 142.0 ± 23.3 pounds. After 12 weeks of exercising three days per week, the lifters gained an average of 0.37 pounds and the walkers increased an average of 0.19 pounds. Although both exercise groups gained significantly in body cathexis during the study, the lifters showed significantly more improvement than the walkers. After adjusting for differences in pretest body cathexis scores and age, women in the resistive training group improved 20.7 ± 21.4 points in body cathexis across the exercise intervention, whereas women in the walking group gained 11.9 ± 17.6 points (F = 3.44, p = .034). After adjusting for differences in reported enjoyment of their respective workouts, in addition to baseline body cathexis and age, the female lifters (M = +23.0) showed 2.4 times greater improvement in body

Table 1
Mean (± SD) Strength and Cardiorespiratory Performance Scores for the Lifters and the Walkers Across the 12-Week Exercise Intervention

	Pretest		Posttest*		F	p
	Lifters (n=30)	Walkers (n=30)	Lifters (n=30)	Walkers (n=30)		
Absolute Strength						
Total (Kg)	152.3 ± 24.5	151.5 ± 21.8	168.2 ± 21.5	158.9 ± 21.1	25.3	.0001
Bench Press	28.3 ± 4.9	28.9 ± 5.1	30.9 ± 4.8	28.9 ± 5.1	20.4	.0001
Leg Press	85.0 ± 15.5	84.1 ± 13.9	94.2 ± 14.7	89.9 ± 12.4	6.2	.0156
Military Press	20.2 ± 4.0	20.2 ± 3.4	22.7 ± 3.1	20.9 ± 3.6	20.6	.0001
Arm Curl	18.8 ± 3.0	18.4 ± 2.6	20.4 ± 2.7	19.1 ± 2.4	12.6	.0008
Dynamic Strength						
Total (Reps)	6.2 ± 7.4	5.0 ± 6.8	12.6 ± 11.4	8.8 ± 10.2	9.5	.0032
Bench Press	3.2 ± 4.4	2.4 ± 3.2	5.0 ± 6.0	3.7 ± 3.9	8.9	.0042
Leg Press	3.1 ± 4.4	2.6 ± 4.1	7.6 ± 6.8	5.0 ± 6.8	6.1	.0169
One Mile Walk						
Time (Sec)	910.6 ± 65.7	957.8 ± 76.2	886.0 ± 70.1	854.5 ± 94.8	3.9	.0540

Note: The F ratios and corresponding p values reflect the extent to which posttest means differ between the lifters and the walkers with the pretest scores controlled.
 * Posttest means have been adjusted for pretest differences.
 Dynamic strength on the bench press was determined using 45% of body weight, whereas dynamic leg press strength was assessed using 140% of body weight with knees flexed at 55°.

cathexis compared to the walkers ($M = +9.6$; $F = 7.3$, $p = .005$).

Not all of the exercisers improved equally in body image across the three-month exercise program. Among the women in the resistive training group, a four variable model emerged to predict improvements in body image accounting for 54.3% of the variance in body cathexis change, as shown in Table 2. In brief, lifters who improved the most in body image were those who: 1) gained the most in absolute muscular strength, 2) began training with the lowest body cathexis scores, 3) reported the most positive feelings at the conclusion of each resistive workout, and 4) indicated they did not plan to participate in forms of exercise other than lifting and/or walking to stay in shape.

Among the women in the walking

group, a five variable model accounted for 54.6% of the variance in body cathexis improvement. Walkers who improved most in body cathexis were those who: 1) recorded the highest training heart rates during their walking workouts, 2) were tallest and 3) were oldest among the walkers, 4) had the lowest body cathexis scores at the onset of the intervention, and 5) had the highest perceived intensity scores at the conclusion of the 12-week walking program.

During the study, both groups of subjects were assigned to exercise on 36 different days. Lifters averaged 31.6 ± 4.3 training sessions and the walkers did not differ significantly averaging 31.2 ± 4.1 workouts ($F = 0.11$, $p = 0.738$).

Regarding participants' subjective ratings assessed using seven-point

Likert scales at the conclusion of the study, lifters and walkers did not differ regarding perceived current fitness level ($M = 4.5 \pm .90$ vs. $M = 4.6 \pm .93$; $F = 0.08$, $p = .779$), perceived improvement in physical fitness across the 12-week intervention ($M = 4.6 \pm 1.5$ vs. $M = 4.6 \pm 1.1$; $F = 0.01$, $p = .922$), nor perceived intensity of their workouts ($M = 4.7 \pm 1.2$ vs. $M = 5.0 \pm 1.1$; $F = 1.42$, $p = .289$), respectively. However, walkers ($M = 5.3 \pm 1.0$) were found to have significantly higher mean ratings regarding the extent to which they enjoyed their workouts compared to the lifters ($M = 4.1 \pm 1.6$), at the conclusion of the study ($F = 11.55$, $p = .0012$).

Lifters (96.7%) and walkers (100%) did not differ regarding the percentage reporting plans to continue exercising ($\chi^2 = 1.0$, $p = .313$), but far more lifters (63.3%) than walkers (26.7%) indicated they planned to weight train in the future to keep fit ($\chi^2 = 8.1$, $p = .004$). Similarly, more walkers (96.7%) than lifters (80.0%) reported they planned to walk to keep in shape ($\chi^2 = 4.0$, $p = .044$).

DISCUSSION

From the results of the present study and numerous other investigations,³⁴ it is clear that regular exercise favorably and significantly improves physical fitness in middle-aged women. As expected, regular resistive training significantly increased muscular strength, dynamic and absolute, and regular exercise walking significantly improved cardiorespiratory endurance across the three-month training period.

Psychological change, particularly improvement in body image, was also associated with participation in the resistive and walking interventions. Of particular interest, participation in the resistive training program resulted in greater increases in body cathexis than involvement in the walking program, consistent with the findings of Trujillo¹⁴ and the suppositions of Tucker.^{12,13}

Table 2

Predictors of Body Cathexis Change Among the Lifters

Source	df	Sum of Squares	F	R ²	p
Model	4	7221.3	7.44	.543	.0004
Strength Change	1	1602.1	6.60	.121	.0165
Pretest Body Cathexis	1	1577.8	6.50	.119	.0173
Workout Rating	1	2439.2	10.05	.184	.0040
Other Exercise Plans	1	1602.2	6.60	.121	.0165
Error	25	6065.7			
Total	29	13287.0			

Workout Ratings = mean rating reflecting the extent subjects enjoyed each workout session

Other Exercise Plans = whether or not subjects planned to engage in regular exercise, other than lifting or walking, in the future

Table 3

Predictors of Body Cathexis Change Among the Walkers

Source	df	Sum of Squares	F	R ²	p
Model	5	4904.2	5.77	.546	.0012
Exercise heart rate	1	1202.4	7.07	.134	.0138
Height	1	866.5	5.09	.096	.0334
Age	1	1004.9	5.91	.112	.0229
Pretest Body Cathexis	1	1141.9	6.71	.127	.0160
Perceived Intensity	1	688.5	4.05	.077	.0556
Error	24	4083.0			
Total	29	8987.2			

Exercise Heart Rate = mean training heart rate of the walkers across the 12-week study

Perceived Intensity = self-reported rating at the conclusion of the study reflecting how intensely subjects felt they trained during the intervention

Resistive training may have caused greater improvements in body image than walking because lifting tends to result in more overt changes in the body than does walking.³⁴ Regular lifting causes muscles to become more toned and defined, whereas the consequences of regular exercise walking tend to be more cardiovascular and less conspicuous. Changes in the body, such as firmer and more shapely muscles, real or imagined, seem to serve as mechanisms of positive feedback which bolster esteem and enhance psychological well-being, particularly body cathexis.^{12,13,16-19,26}

Similarly, performance improvement or success in resistive training may be more noticeable than achievement in walking. Resistive training typically involves multiple exercises, which, if practiced regularly, result in improved muscular strength and endurance in multiple muscle groups. The increased muscular fitness is usually noticed during workouts and especially during strength testing because more resistance can be lifted or more repetitions can be performed than before. Additionally, strength increases are often recorded by lifters, as they were in the present study, and these increases are viewed as successes.

Performance changes in walkers tend to be less numerous and less noticeable than in lifters. Although walking speed tends to increase as cardiorespiratory fitness increases, faster walking times probably invoke less positive feedback than progress in multiple resistive training exercises and tests. In short, lifting more weight and performing more repetitions on the bench press, military press, arm curl, squat, and upright row, probably elicits greater feelings of achievement than walking the mile faster.

Body Cathexis Improvement in the Lifters. The etiological roots underlying improvement in body image experienced by the lifters and the walkers were examined in the present study. As shown in Table 2, among the women who trained

resistively, four variables were predictive of improvements in body image: strength gains, low baseline body cathexis scores, positive workout ratings, and no plans to participate in exercise activities other than lifting or walking.

Evidently, as revealed in a similar study conducted by Tucker in 1983 using male subjects,¹³ those who train the hardest and gain the most in muscular strength during a resistive training program, tend to reap the greatest psychological benefits, particularly regarding body image. This finding suggests that effort and active participation are important elements in the quest to enhance emotional well-being by way of resistive training. Apparently, improvement in body image is limited for those who do not train diligently and therefore fail to gain significantly in muscular strength.

On the other hand, because muscular strength was measured through the use of performance-based tests rather than physiologically-defined measures in the present study, another explanation of the strength change/body cathexis change association warrants consideration. Research shows that performance-based fitness tests tend to be sensitive to motivation-related variables and this is particularly true for muscular strength.³⁵ Thus, the observed gains in dynamic and absolute muscular strength in the lifters may have resulted from the interaction between highly motivated "believers" in the benefits of resistive training and the testing setting, rather than the initial argument. In other words, it is possible that the strength gains observed in the lifters were a function of the psychological alterations experienced by the lifters, rather than the improvements in body image resulting from the changes in muscular strength and related positive feedback, as proposed.

The link between baseline body cathexis scores and body cathexis change indicates that women with relatively low body cathexis scores at the onset of a lifting program tend to improve most in body image from pretest to posttest. However, the

relation could be a function of the "ceiling effect" inherent in self-report, Likert-type inventories.¹³ Women who begin a weight training program with poor body images have no direction to move but up on the posttest, whereas women who begin with high body image scores have little room to improve because of the "ceiling" built into the instrument. Consequently, women with relatively low body image scores at the onset may show the most improvement in body cathexis after participation in a resistive training program.

At the conclusion of each exercise bout, subjects rated their overall feelings on a Likert scale ranging from excellent to very poor. The better lifters felt at the conclusion of each resistive training session, the more they tended to gain in body image across the three-month program. Evidently, participation in weight training workouts that are perceived favorably and are relatively enjoyable is linked to improvements in body image. Again, directionality may be a problem since the opposite is also possible. As body image improves, women may enjoy their workout sessions more.

The last predictor of the four-variable model to explain improvement in body image among the lifters was no future plans to engage in exercise activities other than lifting and/or walking. This finding indicates that lifters with the greatest gains in body image were relatively content with their lifting activities and the program followed by the walkers. Conversely, the lifters who gained relatively little in body cathexis tended to report plans to engage in exercise activities other than resistive training and/or walking. Apparently, these lifters were less satisfied with the exercise programs they witnessed during the previous three months and felt their fitness needs could best be met by engaging in other forms of exercise.

Body Cathexis Improvements in the Walkers. Among the walkers, five measures were predictive of improvement in body cathexis across the 12-week exercise program. Those who

gained the most in body image tended to report higher training heart rates, be taller and older, have lower pretest body cathexis scores, and report higher perceived intensity scores at the conclusion of the study.

Because exercise intensity is directly related to exercise heart rate, it seems that the walkers who trained hardest or most intensely gained the most in body cathexis. As revealed among the lifters, hard work and earnest effort appear to be important ingredients in the pursuit of improved body image in walkers. This same concept was manifest by the fifth variable in the regression model, perceived intensity. Only the exercise heart rate variable was more objective than the perceived intensity measurement. Clearly, both predictors reflect the importance of intensity and hard work in a walking program to enhance psychological well-being.

Taller walkers gained significantly more in body cathexis during the intervention than their shorter peers. Post hoc analyses showed that the taller walkers walked faster during training and performed best on the One-Mile Walk Test. Hence, these walkers probably received the greatest positive feedback due to their superior walking performances and thus experienced the greatest gains in body cathexis.

Older walkers gained significantly more than their counterparts in body cathexis during the study. Post hoc analyses indicated that as age increased in the walkers, improvement on the One-Mile Walk Test increased. Since measured improvement equates to success and affords positive feedback, it is consistent that the older walkers gained more than the younger walkers in body image across the exercise program.

Just as lifters with relatively low pretest body cathexis scores improved most in body cathexis from pretest to posttest, walkers did likewise. It is plausible that the walking program psychologically benefited the women with relatively low body cathexis scores most, but it is equally likely that the inverse

relation was at least partly a function of the "ceiling effect," as discussed earlier.

Conclusions resulting from this study are limited to some extent because a passive control group was not used. Lack of a passive control group did not affect the conclusion that resistive training appears superior to exercise walking for the improvement of body image in middle-aged women. However, without passive controls, it would be inappropriate to conclude that all of the changes that occurred across the exercise intervention were a function of the training programs. For example, some of the changes in body image could have resulted from the Hawthorne Effect.

In summary, it seems that participation in a three-day-per-week resistive training program results in significant gains in muscular strength in middle-aged women, whereas participation in a three-day-per-week walking program leads to significant gains in cardiorespiratory fitness. Furthermore, resistive training seems to promote greater body image improvements than exercise walking in middle-aged women. Additionally, because of the multivariate predictor models unveiled in the present study and the findings of similar research,^{12,15,17} it is clear that women do not improve equally in body image as a consequence of regular resistive training or walking. Evidently, active participation and hard work are important ingredients for improvement in physical fitness and body image among middle-aged, female exercisers.

SO WHAT? Implications for Health Promotion Practitioners

1. This study seems to indicate that regular participation in a three-day-per-week resistive training program enhances body image in middle-aged women more than participation in a three day per week walking program. For lifters and walkers, hard work and increases in fitness seem important for improvements in body image. Further, the present study confirms other fitness research which indicates that middle-aged women can improve substantially in muscular strength by resistive training or in cardiorespiratory endurance by walking.
2. Based on these results and the findings of similar research, practitioners should consider resistive training a tool for the potential enhancement of body image as well as a proven method for the improvement of muscular fitness. Further, practitioners should bear in mind that improvement in body image resulting from exercise tends to occur most in those who train diligently and improve in fitness.
3. Given the findings of this and similar investigations, in future studies, researchers should seek not only to determine *if* exercise improves psychological well-being, but researchers should design studies to confront the important issues of *who* tends to benefit most from *what* types of activities and *why*.

References

1. ACSM. The recommended quality and quantity of exercise for developing and maintaining cardiorespiratory and muscular fitness in healthy adults. *Medicine and Science in Sports and Exercise*, 1990; 22:265-274.
2. Dishman, R. Medical psychology in exercise and sport. *Medical Clinics of North America*, 1985; 69:123-143.
3. Folkins, C., Sime, W. Physical fitness training and mental health. *American Psychologist*, 1981; 36:373-389.
4. Tucker, L. Physical fitness and psychological distress. *International Journal of Sport Psychology*, 1990; 21:185-201.
5. Tucker, L., Cole, G., Friedman, G. Physical fitness: A buffer against stress. *Perceptual and Motor Skills*, 1986; 63:955-961.

6. Brown, D. Exercise, fitness and mental health. In: *Exercise, Fitness and Health: A Consensus of Current Knowledge*. C. Bouchard, R. Shepard, T. Stephens, J. Sutton, B. McPherson (Eds.). Champaign, Illinois: Human Kinetics, 1990; 607-626.
7. Brown, D., Wang, Y. The relationships among exercise training, aerobic capacity, and psychological well-being in the general population. *Medicine, Exercise, Nutrition and Health*, 1992; 1:125-142.
8. Dishman, R. Mental health. In: *Physical Activity and Well-being*. V. Seefeldt (Ed.). Reston, Virginia: AAPERD, 1986; 304-341.
9. Morgan, W., O'Connor, P. Exercise and mental health. In: *Exercise Adherence: Its Impact on Public Health*. R. Dishman (Ed.). Champaign, Illinois: Human Kinetics, 1987; 91-121.
10. Morgan, W., Goldston, S. (Eds.). *Exercise and Mental Health*. Washington, D.C.: Hemisphere Publishing, 1987.
11. Sime, W. Psychological benefits of exercise training in the healthy individual. In: *Behavioral Health: A Handbook of Health Enhancement and Disease Prevention*. J. Matarazzo, S. Weiss, J. Herd, N. Miller, S. Weiss (Eds.). New York: John Wiley and Sons, 1984; 488-508.
12. Tucker, L., Maxwell, K. Effects of weight training on the emotional well-being and body image of females: Predictors of greatest benefit. *American Journal of Health Promotion*, 1992; 6(5):338-344.
13. Tucker, L. Effect of weight training on self-concept: A profile of those influenced most. *Research Quarterly for Exercise and Sport*, 1983; 54:389-397.
14. Trujillo, C. Effect of weight training and running exercise intervention programs on the self-esteem of college women. *International Journal of Sport Psychology*, 1983; 14:162-173.
15. Dishman, R., Gettman, L. Psychological vigor and self-perceptions of increased strength (abstract). *Medicine and Science in Sports and Exercise*, 1981; 13:73-74.
16. Tucker, L. Effect of a weight training program on the self-concepts of college males. *Perceptual and Motor Skills*, 1982; 54:1055-1061.
17. Tucker, L. Effect of weight training on body attitudes: Who benefits most? *Journal of Sports Medicine and Physical Fitness*, 1987; 27:70-78.
18. Tucker, L. Muscular strength and mental health. *Journal of Personality and Social Psychology*, 1983; 45:1355-1360.
19. Tucker, L. Weight training: A tool for the improvement of self and body concepts. *Journal of Human Movement Studies*, 1983; 9:31-37.
20. Brown, R., Harrison, J. The effects of a strength training program on the strength and self-concept of two female age groups. *Research Quarterly for Exercise and Sport*, 1986; 57:315-320.
21. Holloway, J., Beuter, A., Duda, J. Self-efficacy and training for strength in adolescent girls. *Journal of Applied Social Psychology*, 1988; 18:699-719.
22. International Dance-Exercise Association. Fitness consumer survey. *IDEA Today*, October 1989; 28-31.
23. Sonstroem, R. Exercise and self-esteem: Recommendations for expository research. *Quest*, 1982; 33:124-139.
24. Tucker, L. Obesity, exercise, somatotype, and psychological well-being: A factor analytic study. *Journal of Human Movement Studies*, 1983; 9:125-133.
25. Tucker, L. Physical attractiveness, somatotype, and the male personality: A dynamic interactional perspective. *Journal of Clinical Psychology*, 1984; 40:1226-1234.
26. Tucker, L. Relationship between perceived somatotype and body cathexis of college males. *Psychological Reports*, 1982; 50:983-989.
27. Burns, R. *The Self-Concept in Theory, Measurement, Development and Behavior*. New York: Longman, 1979.
28. Tucker, L. Dimensionality and factor satisfaction of the body image construct: A gender comparison. *Sex Roles*, 1985; 12:931-937.
29. Tucker, L. Internal structure, factor satisfaction, and reliability of the Body Cathexis Scale. *Perceptual and Motor Skills*, 1981; 53:891-896.
30. Tucker, L. The structure and dimensional satisfaction of the body cathexis construct of males: A factor analytic investigation. *Journal of Human Movement Studies*, 1983; 9:189-194.
31. ACSM. *Guidelines for Exercise Testing and Prescription* (4th Ed.). Philadelphia: Lea & Febiger, 1991.
32. Hoeger, W., Hopkins, D. *Lifetime Physical Fitness and Wellness: A Personalized Program*. Englewood, Colorado: Morton, 1985.
33. Allsen, P., Harrison, J., Vance, B. *Fitness for Life*. Dubuque, Iowa: Wm. Brown, 1989.
34. Pollock, M., Wilmore, J. *Exercise in Health and Disease: Evaluation and Prescription for Prevention and Rehabilitation*. Philadelphia: Saunders, 1990.
35. Ikai, M., Steinhilber, A. Some factors modifying the expression of human strength. *Journal of Applied Physiology*, 1961; 16:157-163.

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